

Data Storage & Memory

Let's learn how computers remember things! 🧠💻

What is Computer Memory?

Memory is where we store things! A computer system has a memory just like a human brain.

It stores data and information. Unlike human memory, computer memory keeps data for a long time until you erase it!

Fact: Computer memory is also known as a **Storage Device**.



Cartoon computer with a brain

Types of Computer Memory



Primary Memory

Also known as **Internal Memory** or Main Memory.

It is embedded inside the computer. It stores data that the CPU accesses directly.

Types: RAM & ROM.



Secondary Memory

Also known as **Auxiliary Memory**.

It is external memory. It has a huge storage capacity to store data permanently until deleted.

Examples: Hard Disk, CD, DVD, Pen Drive.

Inside Primary Memory

RAM (Random Access Memory)

- It is a **Read/Write** memory.
- **Volatile:** It forgets everything when the computer is turned off! Data is temporary.

ROM (Read Only Memory)

- It is a **Read Only** memory.
- **Non-Volatile:** It remembers everything perfectly, even when power goes off!

 Cartoon image of a computer RAM chip stick

Secondary Storage Devices

Hard Disk

Enormous storage! Composed of multiple magnetic disks called platters.

Fixed inside the CPU box or available externally.

Compact Disk (CD)

A shiny circular plastic plate. Stores about 700 MB of data like audio, video, or text.



More Storage Devices!

Pen Drive

A USB Flash Drive! Very small in size but can store a large amount of data (4GB to 512GB+).

Easy to carry in your pocket!

Blu Ray Disk

Looks like a DVD but can store huge amounts of data (up to 25GB) and high-definition video!

Uses a blue laser beam to read/write data.



Measuring Computer Memory

Memory Unit	Value
1 Bit	Binary Digit (0 or 1)
1 Nibble	4 bits
1 Byte	8 bits
1 Kilobyte (KB)	1024 Bytes
1 Megabyte (MB)	1024 KB
1 Gigabyte (GB)	1024 MB

Quiz Time! Question 1

What are the two main types of Primary Memory?

A) Hard Disk & CD

B) RAM & ROM

C) Bits & Bytes

✓ Answer: B) RAM & ROM



Quiz Time! Question 2

Which memory is volatile and forgets everything when the power is turned off?

A) ROM

B) Hard Disk

C) RAM

✓ Answer: C) RAM!

 Cute thinking cartoon kid

Quiz Time! Question 3

Which storage device is small, uses USB, and can be easily carried in a pocket?

A) Hard Disk

B) Pen Drive

C) CD

✓ Answer: B) Pen Drive!



Quiz Time! 📝 Question 4

How many bits make up 1 Byte?

- A) 4 bits
- B) 1024 bits
- C) 8 bits

★ Answer: C) 8 bits!



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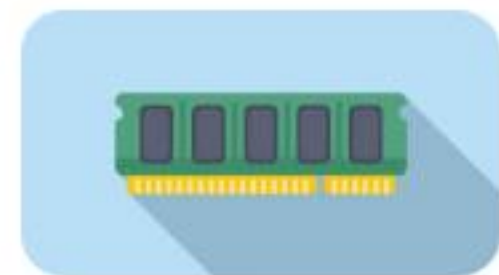


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